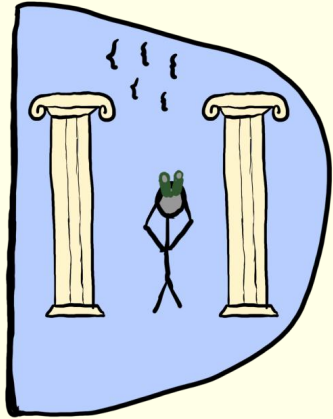


Translating Between the Common Haar State Model and the Unitary Model

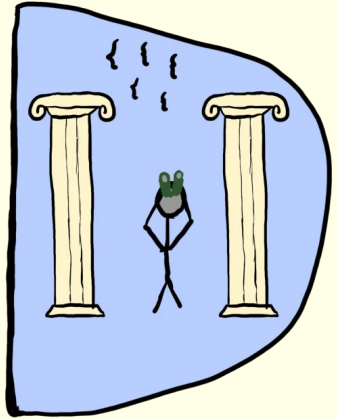
Eli Goldin, Mark Zhandry

Oracle Separations



arbitrary deterministic
function on strings

Oracle Separations



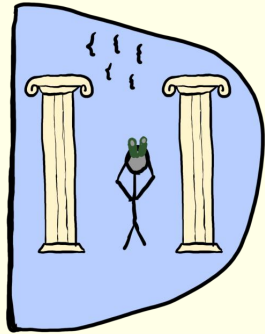
- arbitrary deterministic
function on strings

can be distribution.

e.g. Random Oracle

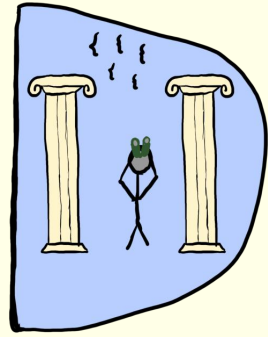
Oracle Separations

Primitive
exists in

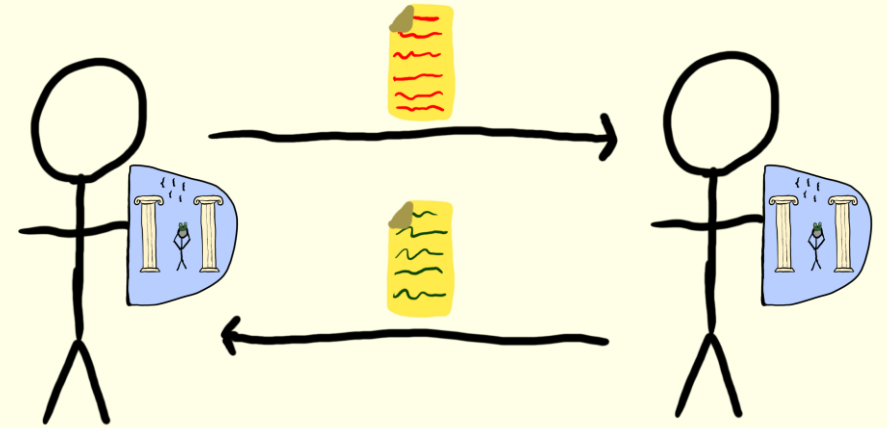


Oracle Separations

Primitive
exists in



if you
have

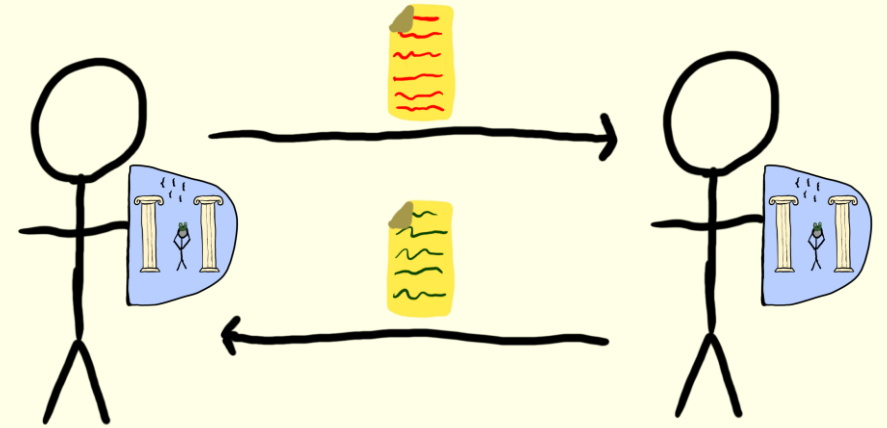
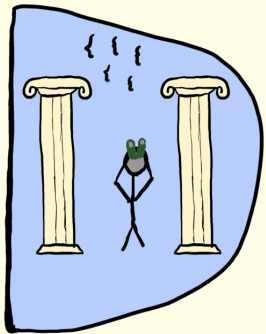


Secure
against

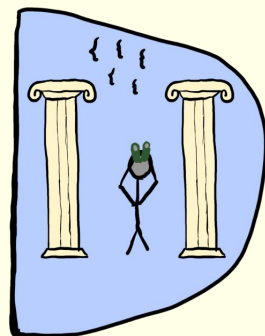


Oracle Separations

"efficient"
= poly calls
to

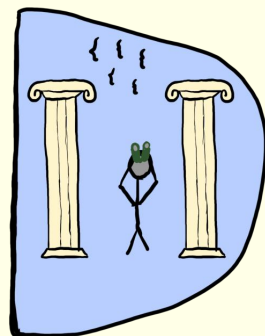


[IR89]



= Random Oracle

[IR89]



= Random Oracle



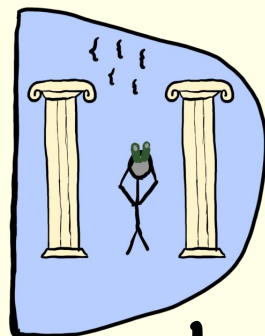
(x)

ONE WAY



OWFs exist

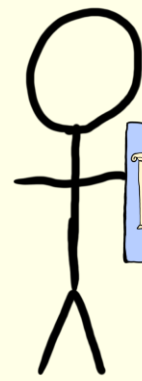
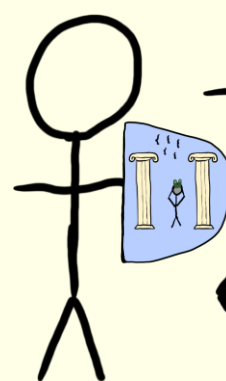
[IR89]



= Random Oracle



(x)



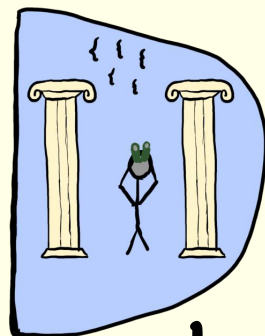
KE does not

ONE WAY



OWFs exist

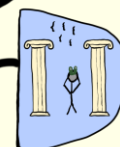
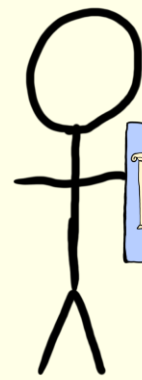
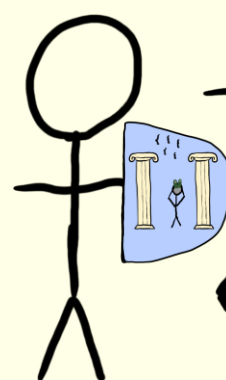
[IR89]



= Random Oracle



(x)



KE does not

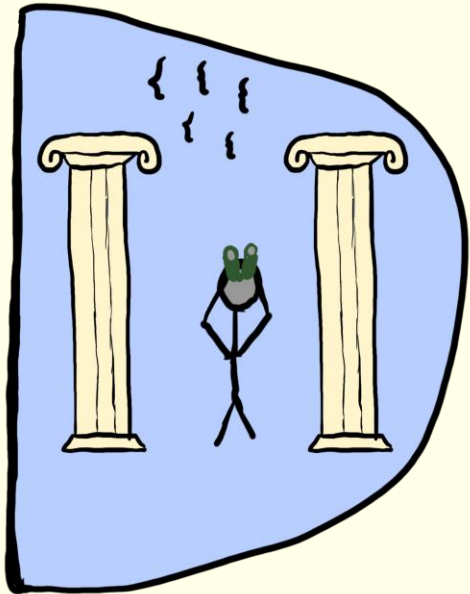
OWFs exist

Cannot build KE from OWF in "black-box" way!

Quantum Oracles

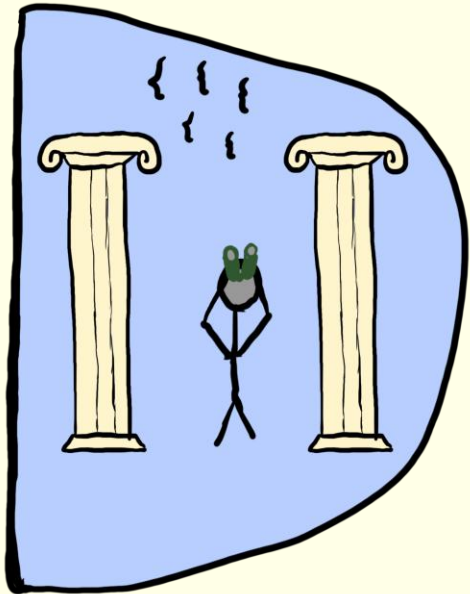
To separate quantum
primitives, need quantum
accessible oracles!

Quantum Oracles - Modeling



(quantum accessible) classical oracle

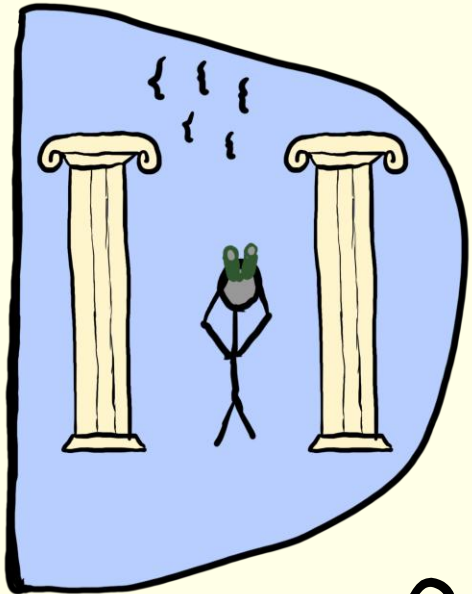
Quantum Oracles - Modeling



(quantum accessible) classical oracle

functionality: any function on strings

Quantum Oracles - Modeling



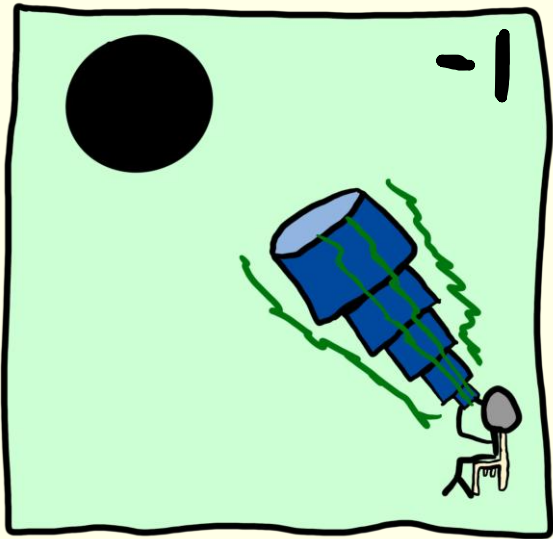
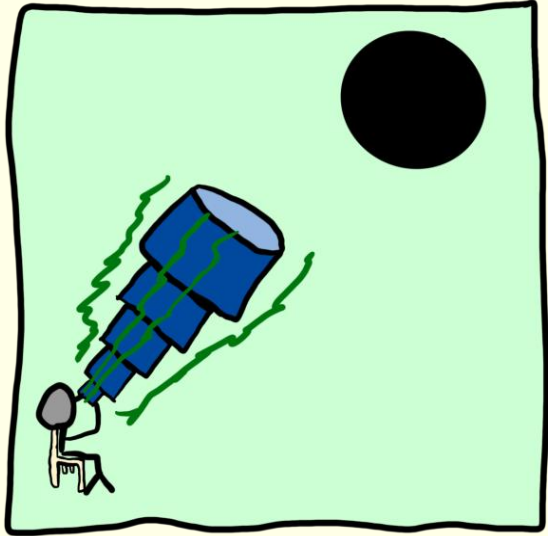
(quantum accessible) classical oracle

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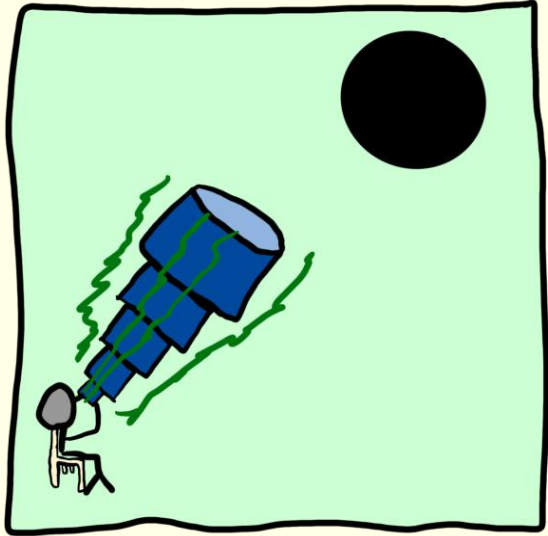
oracle separation rules out any
black-box construction

Quantum Oracles - Modeling

Unitary oracle

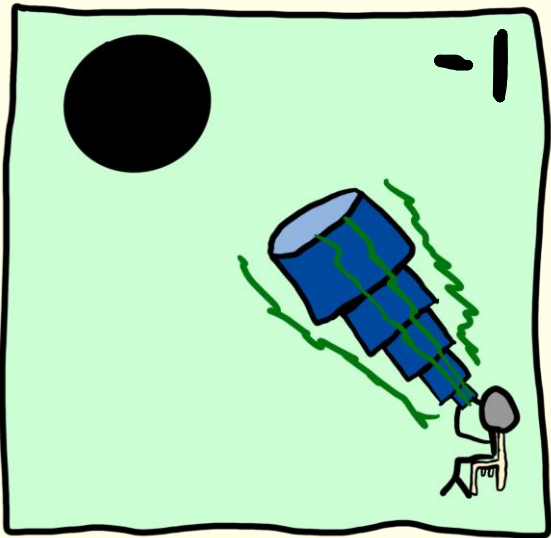


Quantum Oracles - Modeling

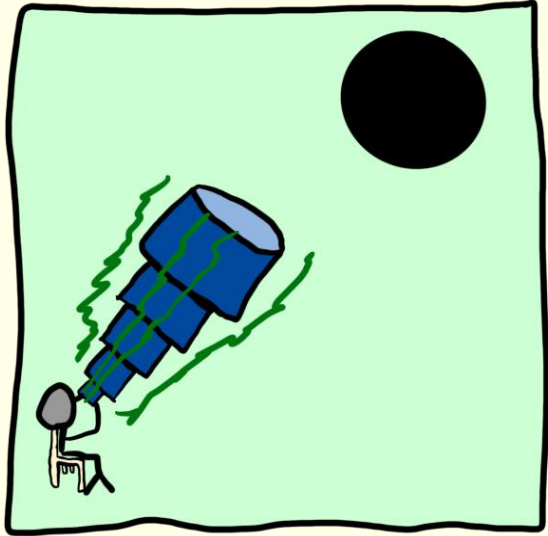


Unitary oracle

functionality: any unitary
operation and its inverse

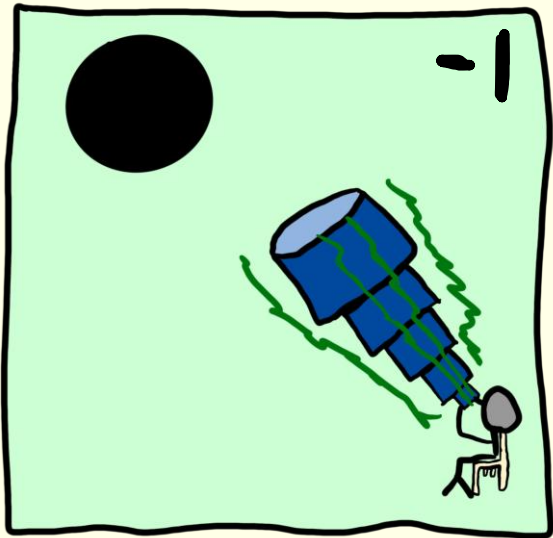


Quantum Oracles - Modeling



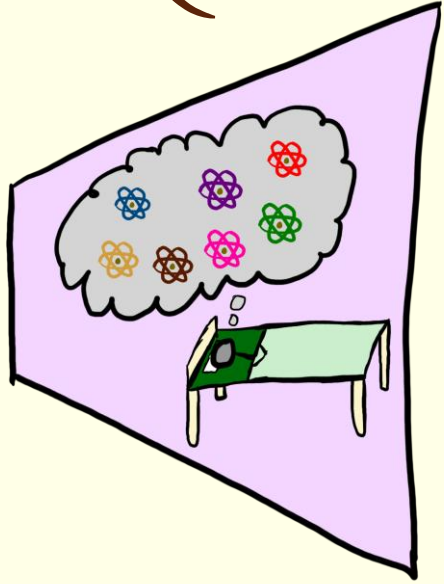
Unitary oracle

functionality: any unitary
operation and its inverse



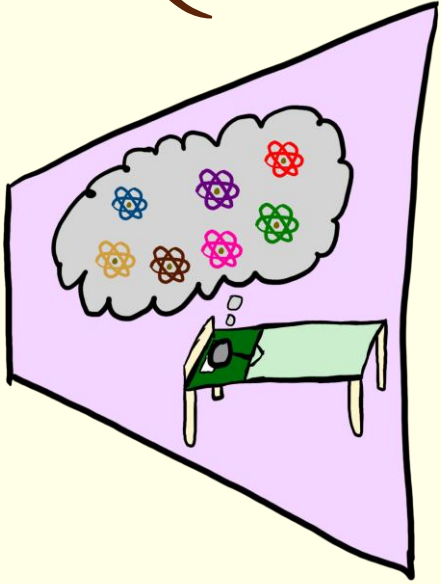
oracle separation rules out
most black-box constructions

Quantum Oracles - Modeling



Isometry oracle

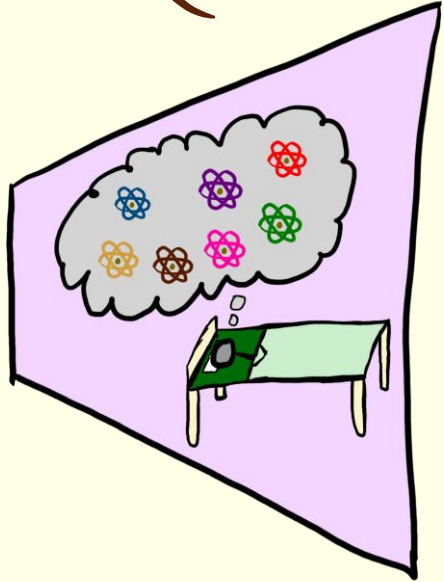
Quantum Oracles - Modeling



Isometry oracle

functionality: any isometry
(irreversible quantum operation)

Quantum Oracles - Modeling

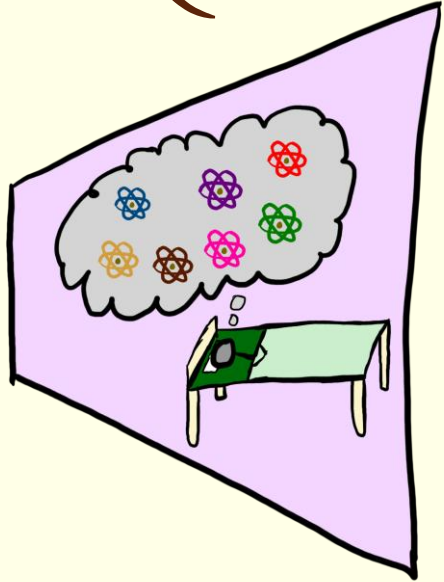


Isometry oracle

functionality: any isometry
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Oracle separation rules out black-box
constructions which do not "run the
primitive in reverse"

Quantum Oracles - Modeling

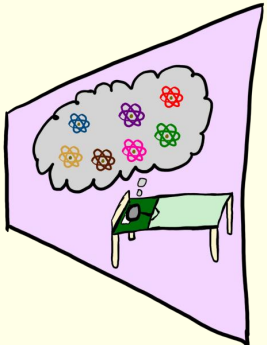
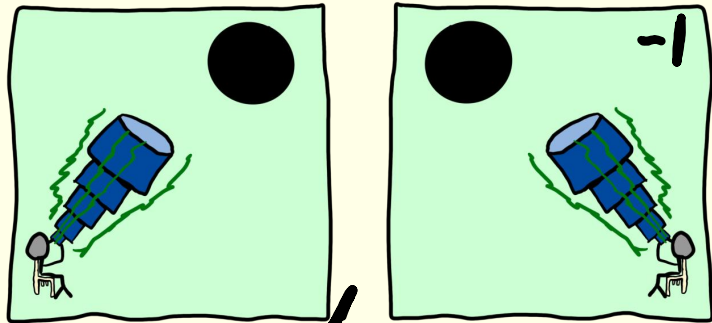
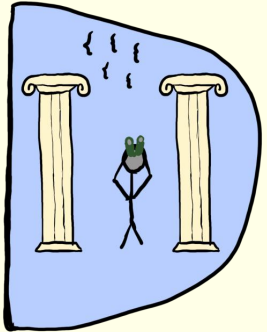


Isometry oracle

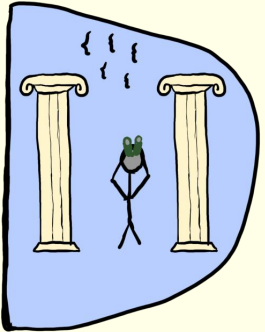
functionality: any isometry
(irreversible quantum operation)

Oracle separation rules out black-box
constructions which do not "run the
primitive in reverse" - fairly common

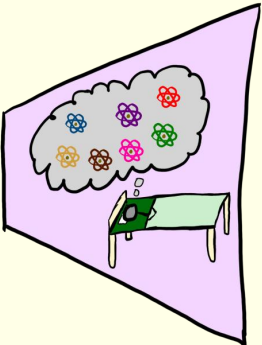
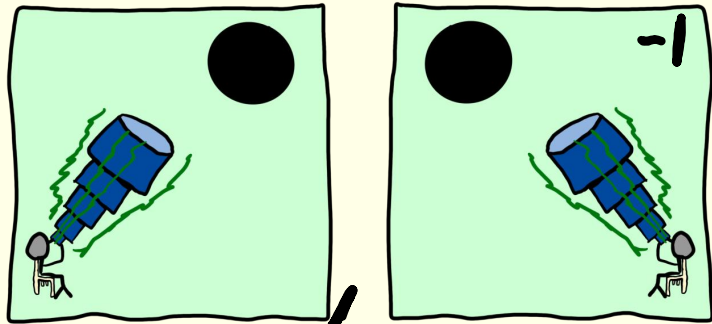
Quantum Oracle Separations - Ranked



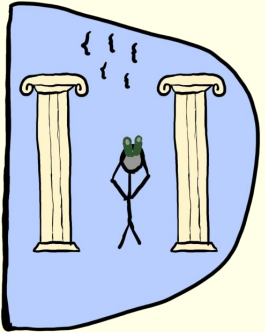
Quantum Oracle Separations - Ranked



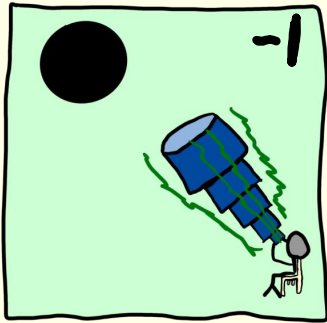
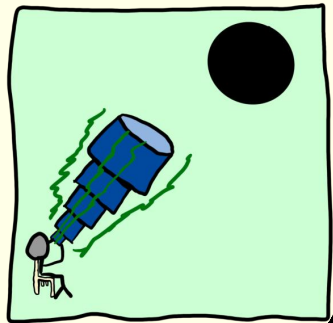
best we can hope for



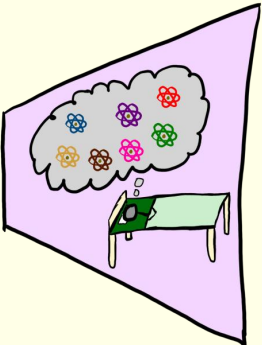
Quantum Oracle Separations - Ranked



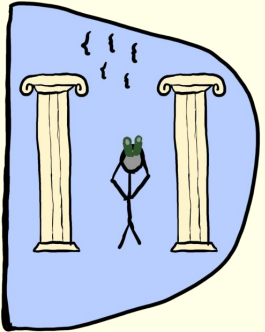
best we can hope for



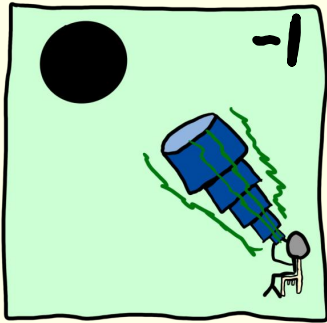
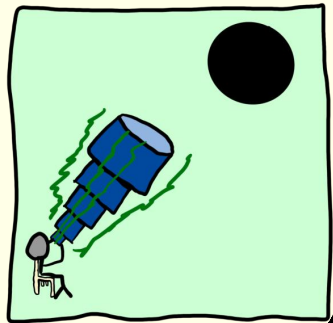
good enough



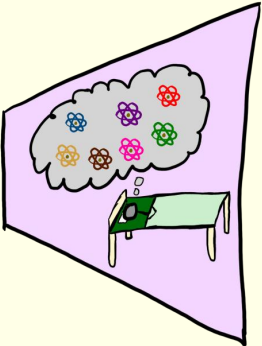
Quantum Oracle Separations - Ranked



best we can hope for



good enough



preliminary result

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$
fixed Haar random state

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

exists

does NOT exist

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

| poly-copy pseudorandom states

exists

| does NOT exist

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

| poly-copy pseudorandom states

| classical communication (QCC)

| cryptography

| does NOT exist

exists

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

| poly-copy pseudorandom states

| classical communication (QCC)

| cryptography

| ...

| does NOT exist

exists

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

1-copy pseudorandom states		poly-copy pseudorandom states
		classical communication (QCC)
		cryptography
		...
exists		does NOT exist

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

1-copy pseudorandom states	poly-copy pseudorandom states
quantum commitments	classical communication (QCC)
	cryptography
	...
exists	does NOT exist

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

1-copy pseudorandom states	poly-copy pseudorandom states
quantum commitments	classical communication (QCC)
...	...
exists	does NOT exist

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

1-copy pseudorandom states | poly-copy pseudorandom states

quantum commitments | classical communication (QCC)

cryptogaphy

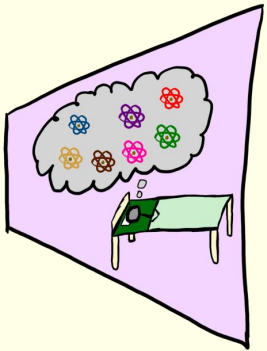
...

exists

does NOT exist

Common Haar Random State Model

CHRS: nothing $\rightarrow |\psi\rangle$

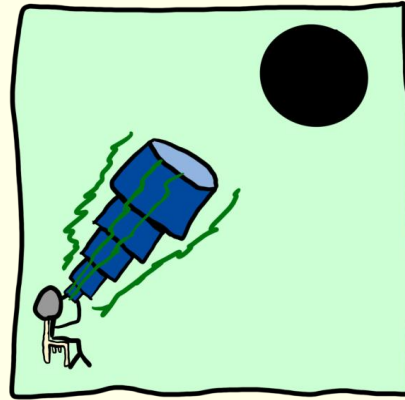


- NOT reversible
isometry oracle

Common Haar Random State Model

Big Question: Can we make these separations unitary?

Common Haar Random State Model

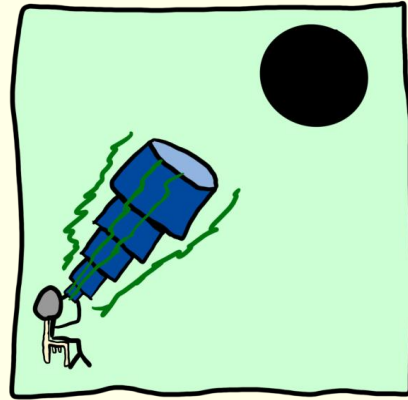


$[CCS25]$

$[BCN25]$

$[BMM^+25]$

Common Haar Random State Model



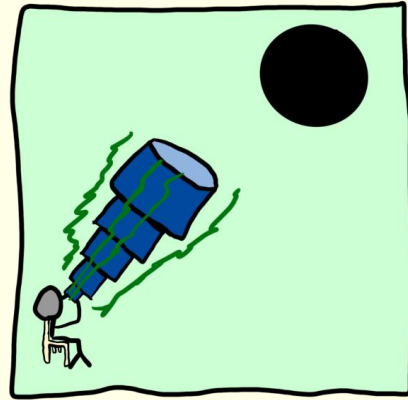
$[CCS25]$

$[BCN25]$

$[BMM^+25]$

SWAP: $|0\rangle \leftrightarrow |4\rangle$

Common Haar Random State Model



$[CCS25]$

$[BCN25]$

$[BMM^+25]$

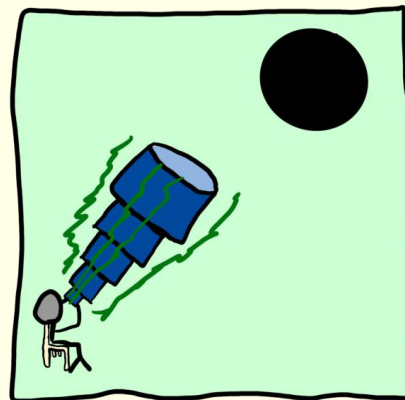
SWAP: $|0\rangle \leftrightarrow |4\rangle$

— — — — — — — — — —

subset swap oracle

Common Haar Random State Model

proof
breaks down :)



~~[CCS25]~~

~~[BCN25]~~

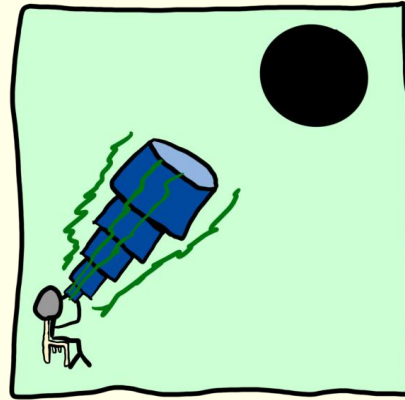
[BMM⁺25]

~~SWAP: $|0\rangle \leftrightarrow |4\rangle$~~

subset swap oracle

Common Haar Random State Model

proof
breaks down :)



~~[CCS25]~~

~~[BCN25]~~

[BM⁺25]

~~SWAP: $|0\rangle \leftrightarrow |4\rangle$~~

subset swap oracle, ✓

Conjecture

CHRS: nothing $\rightarrow 147$

SWAP: $107 \leftrightarrow 147$

Conjecture: all oracle separations in
CHRS also hold in SWAP.

i.e. P exists in CHRS

$\Leftrightarrow P$ exists in SWAP

Our Results

Thm^{*}: P exists in CHRS
 $\Rightarrow P$ exists in SWAP.

Our Results

Thm^{*}: P exists in CHRS
 $\Rightarrow P$ exists in SWAP.

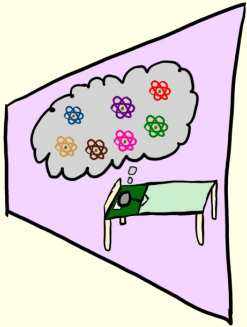
Thm: P exists in SWAP
 $\Rightarrow P$ exists in CHRS-.

Our Results

Thm^{*}: P exists in CHRS
 $\Rightarrow P$ exists in SWAP.

Thm: P exists in SWAP
 $\Rightarrow P$ exists in CHRS-.
similar, most proofs work!

Unitary separation template



CHRS

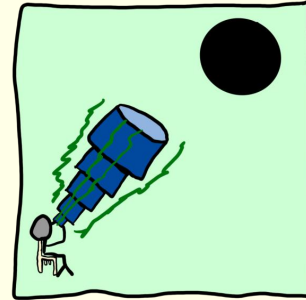
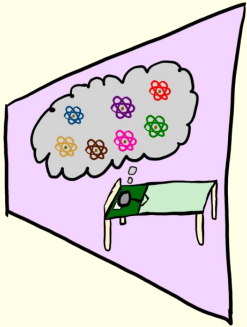
P_1

exists

P_2

does NOT
exist

Unitary separation template



CHRS

P_1

exists

P_2

does NOT
exist

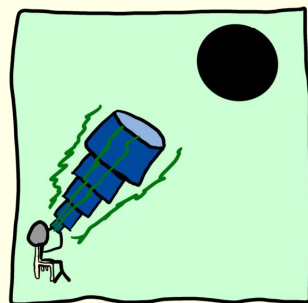
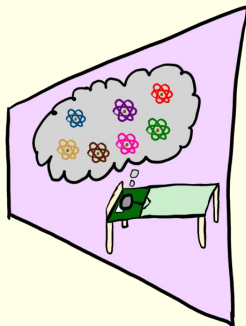
P_1

exists

SWAP

FREE

Unitary separation template



CHRS

P_1		P_2
exists		does NOT exist

P_1
exists

SWAP

P_2
does NOT exist

FREE

adapt proof

CHRS- Model

For $|\psi\rangle$ any state,

define $|\psi-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |\psi\rangle).$

CHRS- Model

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CHRS-: nothing $\rightarrow |\psi-\rangle$

CHRS- Model

For $|\psi\rangle$ any state,

define $|\psi-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |\psi\rangle)$.

CHRS-: nothing $\rightarrow |\psi-\rangle$
↳ fixed Haar random $|\psi\rangle$

Computing SWAP with CRSH-

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$ CRSH: nothing $\rightarrow |\psi\rangle$

Computing SWAP with CRSH-

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$ CRSH-: nothing $\rightarrow |\psi\rangle$

$$\text{SWAP}|\psi\rangle = \text{SWAP}\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|\psi\rangle\right)$$

Computing SWAP with CRSH-

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$ CRSH-: nothing $\rightarrow |\psi\rangle$

$$\text{SWAP}|\psi\rangle = \text{SWAP}\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|\psi\rangle\right)$$

$$= \frac{1}{\sqrt{2}}|\psi\rangle - \frac{1}{\sqrt{2}}|0\rangle = (-1) \cdot |\psi\rangle$$

Computing SWAP with CRSH-

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$ CRSH-: nothing $\rightarrow |\psi\rangle$

SWAP: $|\psi\rangle \leftrightarrow (-1) \cdot |\psi\rangle$

SWAP is reflection around $|\psi\rangle$.

Computing SWAP with CRSH-

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$ CRSH-: nothing $\rightarrow |\psi\rangle$

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SWAP is reflection around $|\psi\rangle$.

[JLS 18, BMM+25]: Given $|\psi\rangle|\psi\rangle\cdots|\psi\rangle$, can
compute reflection around $|\psi\rangle$.

Computing SWAP with CRSH-

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$ CRSH-: nothing $\rightarrow |\psi\rangle$

SWAP: $|\psi\rangle \leftrightarrow (-1) \cdot |\psi\rangle$

SWAP is reflection around $|\psi\rangle$.

[JLS 18, BMM+25]: Given $|\psi\rangle|\psi\rangle\cdots|\psi\rangle$, can
compute reflection around $|\psi\rangle$. DONE

Why?

Thm^{*}: P exists in CHRS
 $\Rightarrow P$ exists in SWAP.

Thm: P exists in SWAP
 $\Rightarrow P$ exists in CHRS-.

similar, most proofs work!

Why?

SWAP

can
compute

CHRS

CHRS-



Why?

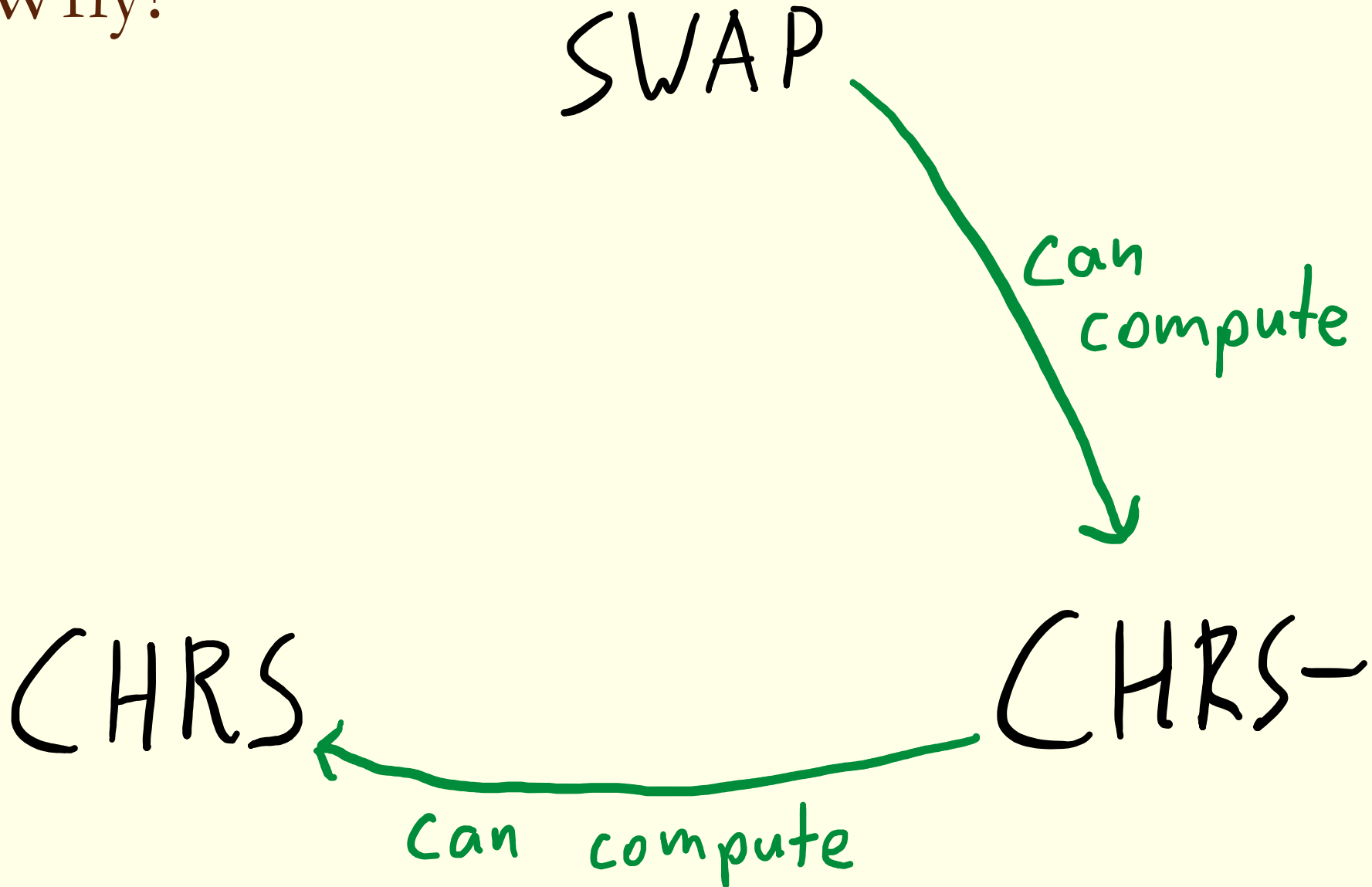
SWAP

can
compute

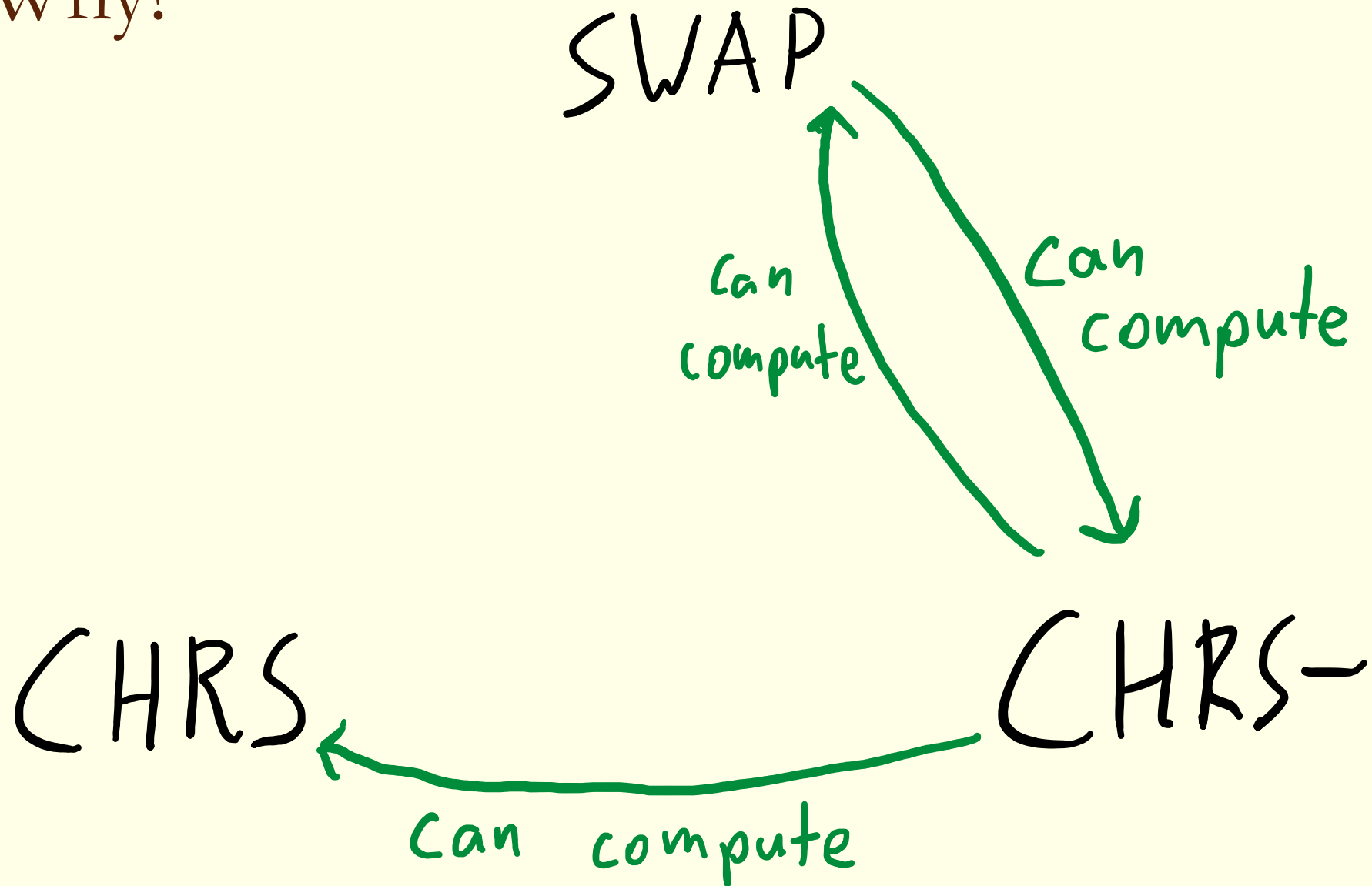
CHRS

CHRS-

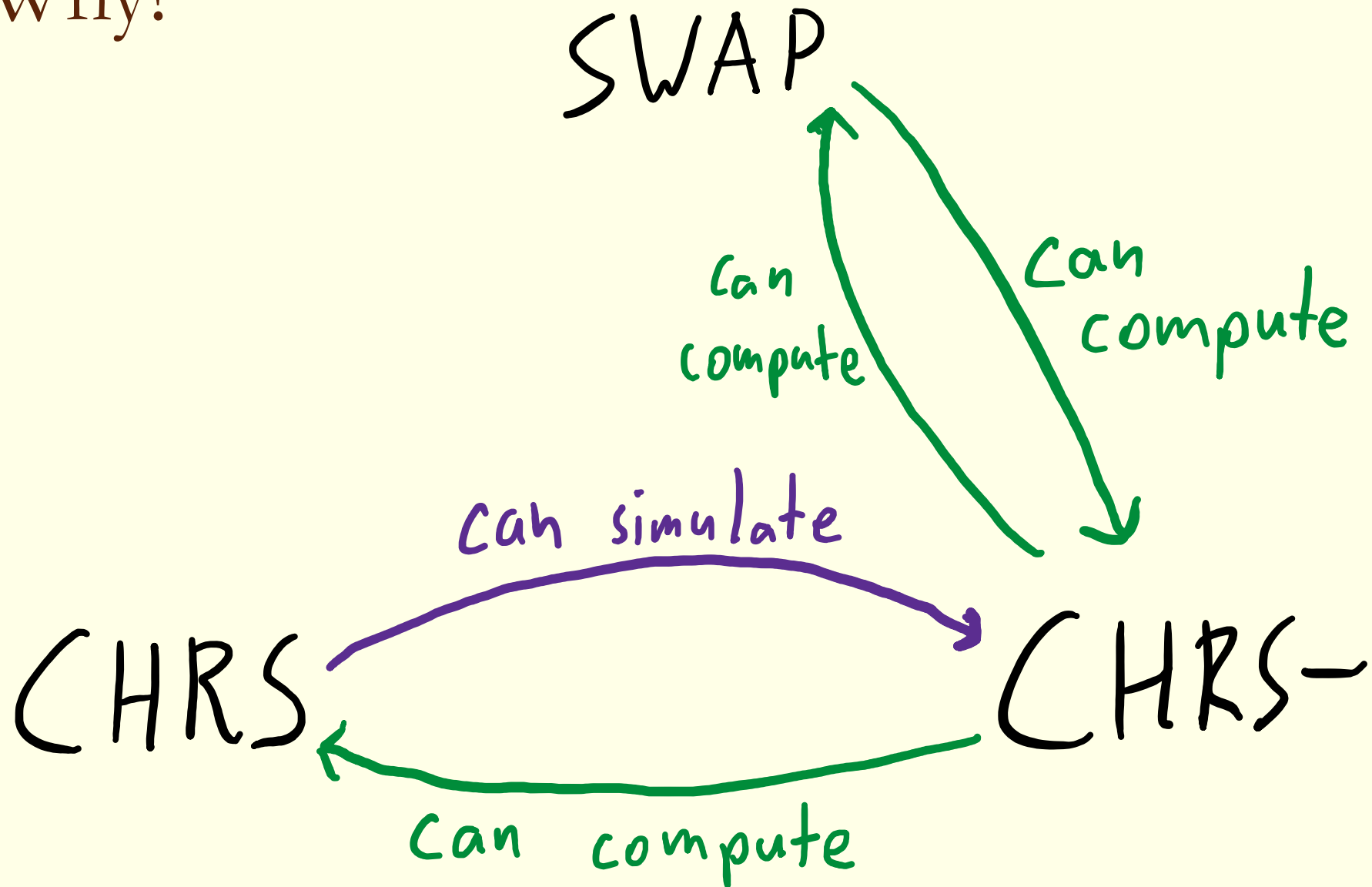
can compute



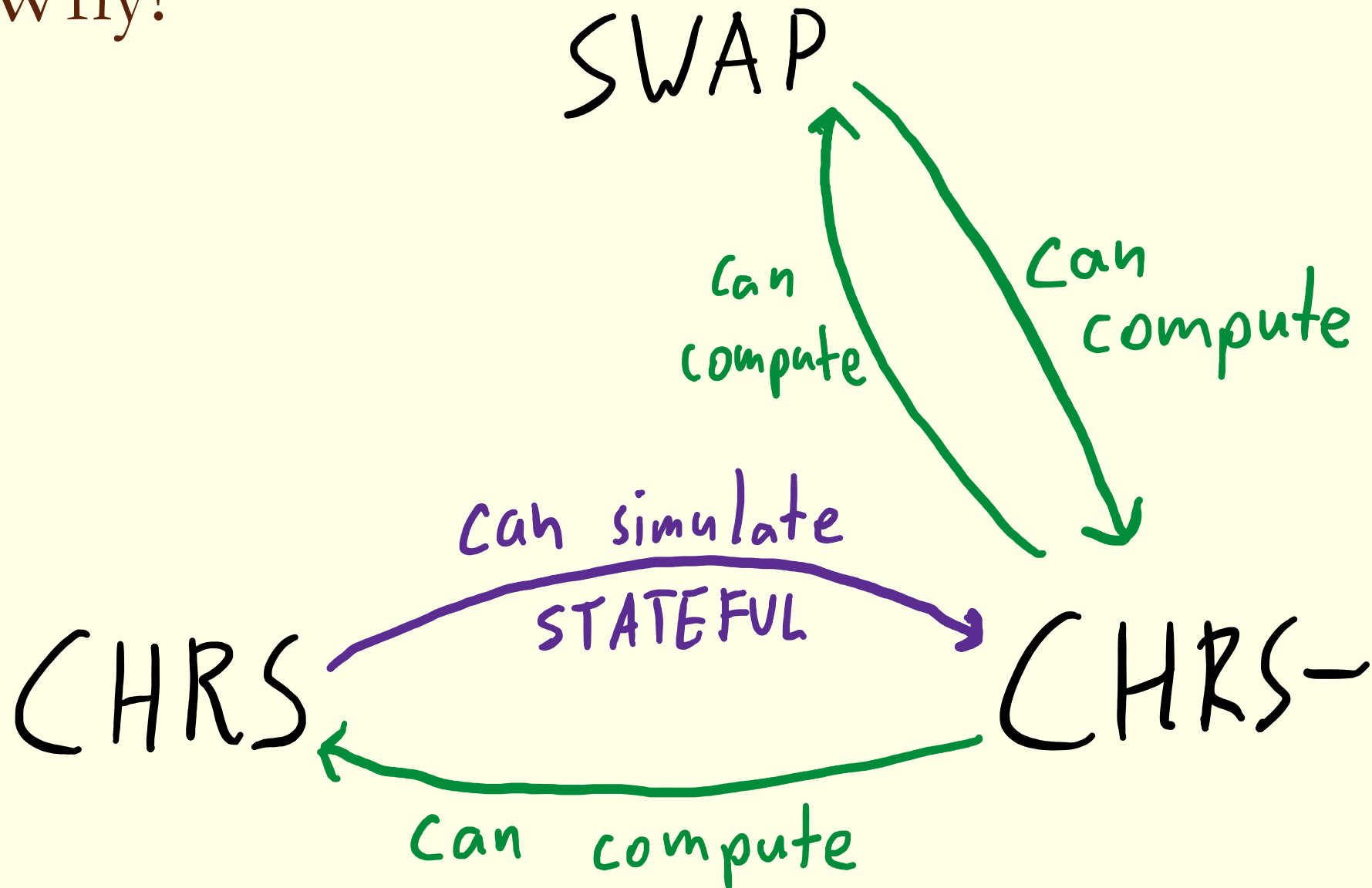
Why?



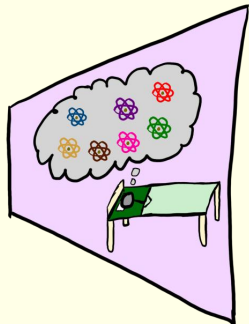
Why?



Why?



Our Results



CHRS: nothing $\rightarrow |\psi\rangle$

1-copy pseudorandom states

quantum commitments

...

exists

poly-copy pseudorandom states

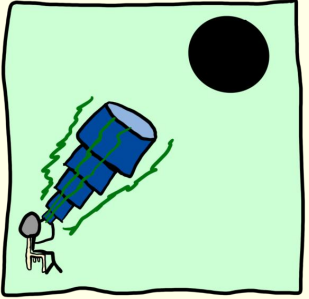
classical communication (QCC)

cryptography

...

does **NOT** exist

Our Results



1-copy pseudorandom states

quantum commitments

...

exists

SWAP: $|0\rangle \leftrightarrow |\psi\rangle$

| poly-copy pseudorandom states

| classical communication (QCC)

| cryptography

| ...

| does NOT exist

Open Questions

P^{CHRS-}

secure in $CHRS-$



$\exists \tilde{P}^{CHRS}$

secure in $CHRS?$

Open Questions

P^{CHRS-}

secure in $CHRS-$



$\exists \tilde{P}^{CHRS}$

secure in $CHRS?$

No candidate!

Open Questions

CHRS $\xrightarrow[\text{STATEFULLY}]{\text{can simulate}}$ CHRS-

Open Questions

CHRS $\xrightarrow[\text{STATEFULLY}]{\text{can simulate}}$ CHRS-

can't be used in
stateless / deterministic / pure
primitives...

Open Questions

P^{CHRS-}

secure in $CHRS-$



$\exists \tilde{P}^{CHRS}$

secure in $CHRS?$

Open Questions

$P^{\text{CHRS-}}$

secure in CHRS-



$\tilde{P}^{\text{CHRS}}$

secure in CHRS?

Relationships between other quantum oracles?

Open Questions

$P^{\text{CHRS-}}$

secure in CHRS-



$\tilde{P}^{\text{CHRS}}$

secure in CHRS?

Relationships between other quantum oracles? Haar unitary, QRDM, ...